27	Two beams of monochromatic light have the same frequency. One beam has a greater intensity than the other beam. How can the greater intensity of one beam be explained in terms of the particulate nature of light?	
	A	Each photon in the beam has more energy.
	B	Each photon in the beam has greater speed.
	C	The amplitude of oscillation is greater.
	D	There are more photons passing through unit cross-sectional area of the beam per unit time.
L1	**Answer: D** Intensity = number of photons per unit time x energy of one photon ÷ cross section area of beam = number of photons per unit time per unit area x energy of one photon Since the two beams have the same frequency, the energy of one photon is the same, option A is not correct. All photons travel at the same speed of light in vacuum. So option B is not correct. Amplitude of oscillations of electric fields and magnetic fields applies to the wave nature of light. So option C is not correct.	